RoboCupJunior Rescue Simulation Competition Platform

For RoboCupJunior Rescue Simulation and similar competitions.

MIT License | Python 3.12 | Django 6.1 | Docker Compose

A web platform that helps organizers run a competition, gives teams a private workspace, and lets the public follow schedules, results and interactive match replays.

Key features

- **Interactive 3D replays** with browser playback, seeking and camera controls.

- **Automatic results** from recording packages, plus flexible scoring rules.

- **Drag-and-drop scheduling** and team lineups.

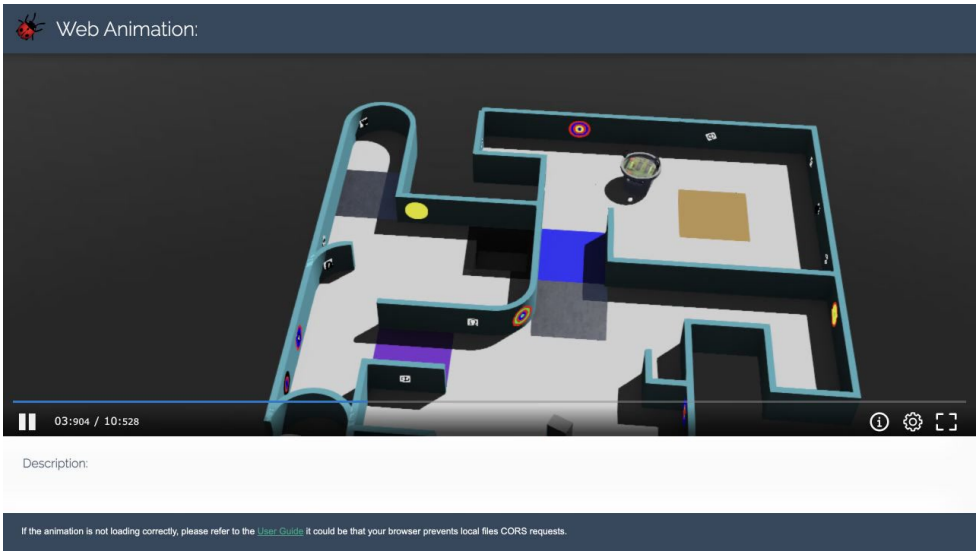
- **Private submissions** with deadlines, upload locking and controlled access.

- **Competition tools** for maps, files, announcements, teams and complaints.

See a demonstration

Demonstrated for the **Junior Rescue Simulation League at RoboCup Americas Super Regional (ASR) 2026**. This is one example of the platform and its features.

View RCJRescue.com | Watch the YouTube demonstration



Actual Erebus/Webots replay footage. Source code is available under the MIT License.

01 / Recording to interactive replay

Upload an Erebus/Webots folder or ZIP. Assets retain their names and relative paths in an isolated package. A preview and Play button open a responsive popup with the exported player, seeking, camera navigation and fullscreen. A video can also be supplied.

The recording metadata fills score and time only where those fields are empty. Final score already includes bonuses. Existing manual values are preserved. Exported event scripts follow the playback timeline rather than showing the final event throughout the run.

02 / Round lineup to calendar

Arrange the teams within a round. Place its block in the calendar and individual run times follow the lineup. Drag an event to move it, click to edit in a popup, or select an empty interval to add an activity.

03 / Scores to reviewed standings

Enter scores in a team-by-round matrix, paste from a spreadsheet, undo edits and spot missing values. Main and SuperTeam score tables appear together. Rank publication requires an administrator decision even when scores are already public.

Default field calculation: normalize each round against its highest raw score, discard each team's configured number of lowest normalized scores, then average the retained scores and multiply by 100. Keep at least one round. Exact ties remain ties unless a tie-break is configured; zero-maximum rounds require an organizer decision.

04 / Controlled access and communication

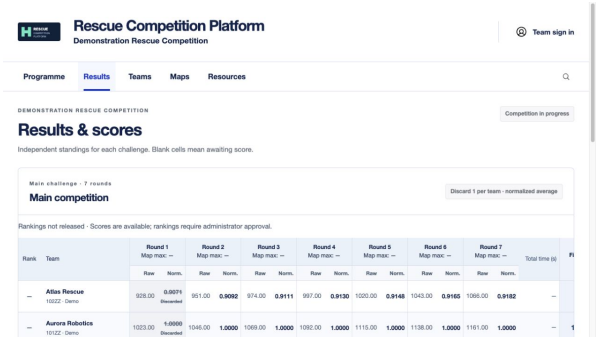
Named quarantine requests support team-specific deadlines and private multi-file submissions. Organizer packages and announcements use one record with a recipient list. Search, select all and clear controls make audience selection direct. Removing a recipient removes permission.

Additional tools include map-folder detection, metadata and GLB views; SuperTeam pairings using existing teams; complaint attachments and organizer responses; account management; optional personalized email; storage configuration and activity records.

Missing recording metadata does not become a zero result. Older recordings remain playable. Public release and team privacy remain separate decisions.

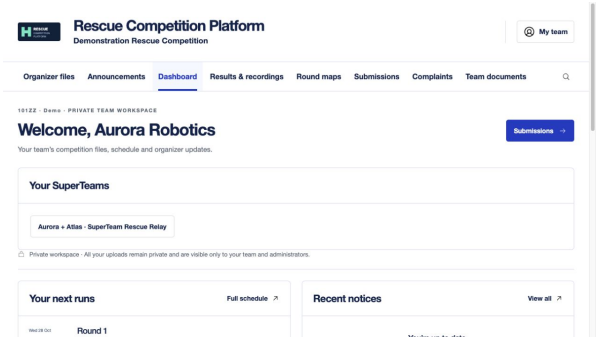
Public view / Follow the competition

Schedules, public announcements, released scores and rankings, interactive match replays and released 3D maps. Watch, seek and rotate a match in a browser without installing Webots.



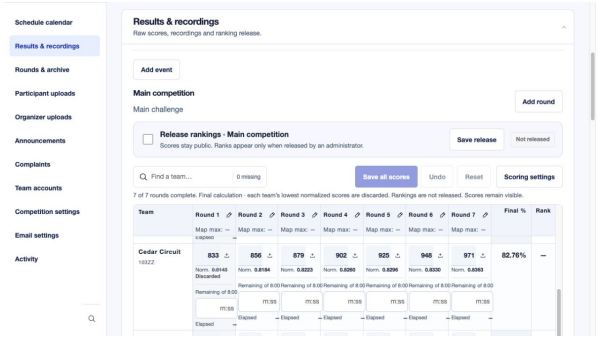
Team view / A private workspace

Upcoming runs, targeted resources and announcements, private multi-file submissions with deadlines, SuperTeam workspaces, account tools and complaints with attachments.



Admin view / Connected event management

Configure teams, rounds and scoring; arrange the calendar; import recordings to fill empty score and time fields; manage quarantines, maps, recipients and publication. Rankings require explicit release.



Get started

Use Python 3.12 for development, or Docker Compose for the application, PostgreSQL, scanner and background worker. The setup guide provides commands for a fresh installation. Create your own administrator account; no shared account or password is bundled.

The optional demo command creates fictional teams, scores and a SuperTeam challenge. It sends no email and creates no login accounts. Do not run it on a real competition database.

Before operating a real event

Configure the event dates, timezone, challenges and scoring policies. Set up teams, lineups, upload windows and recipients. Configure SMTP and storage outside version control. Use HTTPS, restricted private storage, malware scanning and tested backups. Review release settings before publishing maps, recordings or rankings.

Source preparation and publication

The separate repository uses a new source history. The original database, participant accounts, submissions, recording packages, environment secrets and production operational notes are excluded. Screenshots come from an isolated fictional local demo. The deployed competition has not been changed for this preparation.

Verification: all 358 Python tests passed on PostgreSQL, including concurrency checks, plus all 10 JavaScript checks. Production checks and dependency and secret scans passed. Passing checks are not a guarantee of complete security.

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Documentation and video

The README includes the feature catalogue, screenshots, quick start and a linked video preview. Setup, deployment, security and contribution guides are included. The public walkthrough has English captions and chapter links.

Watch the feature walkthrough by Peyman Yousefi

Public release authorized by Peyman Yousefi on 27 October 2026. Proposed contributions require owner review before merging.